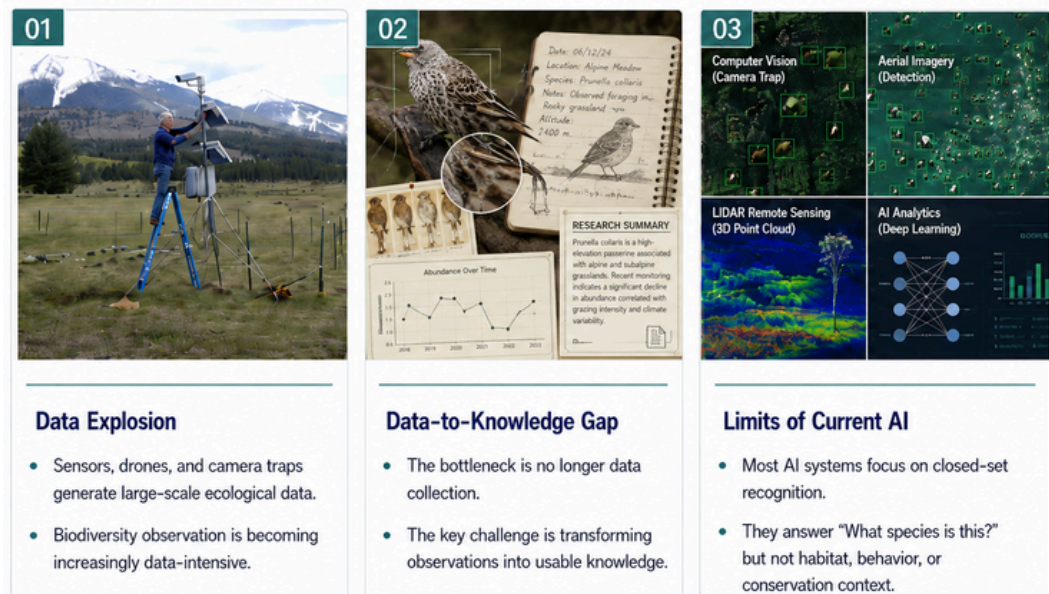


E pluribus : A Multi-Task Vision-Query2Contextual Framework for Ecological Research Assistant

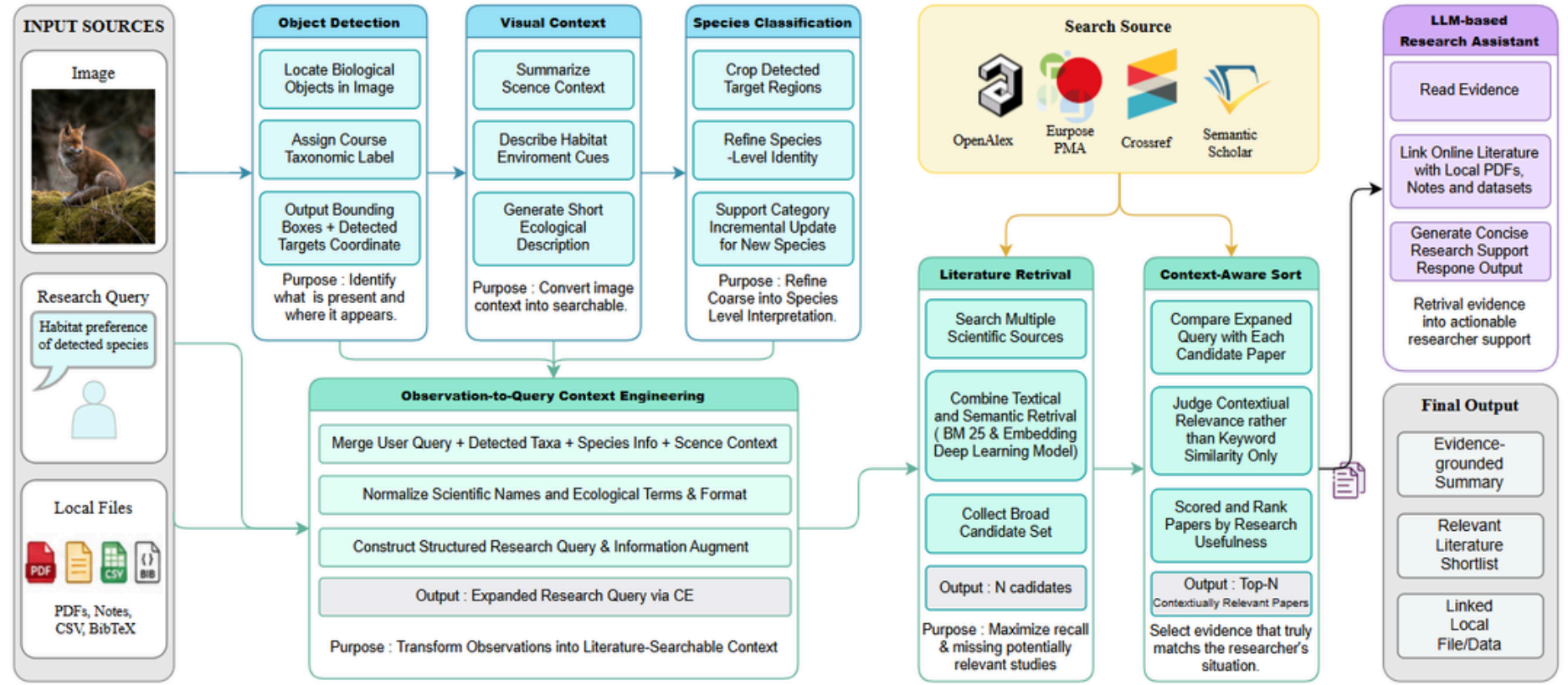
Jia-En Suen, Chun-Hao Chen

Introduction



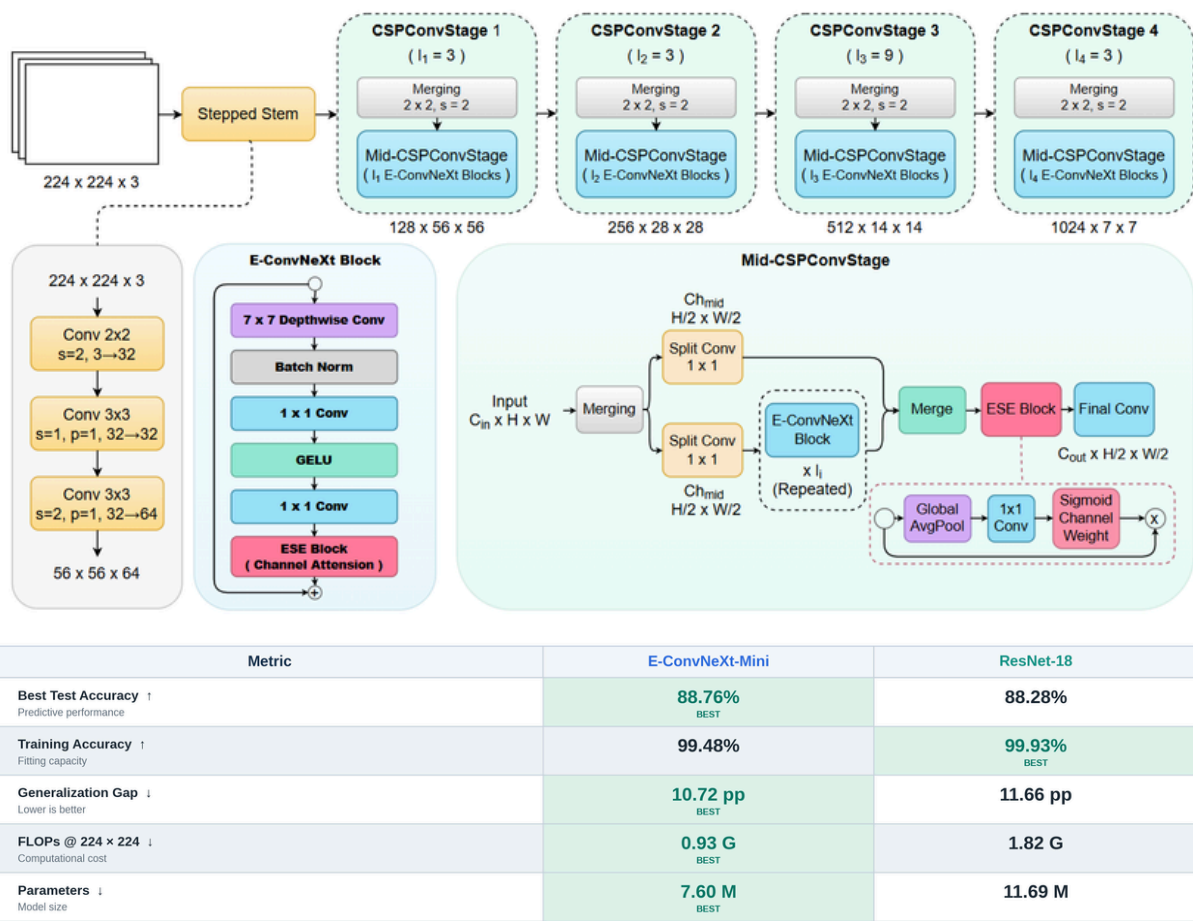
A local AI assistant for ecological research, integrating field images, species identification, literature, and research records. It filters evidence by species, habitat, behavior, and intent to support ecological analysis.

System Overview

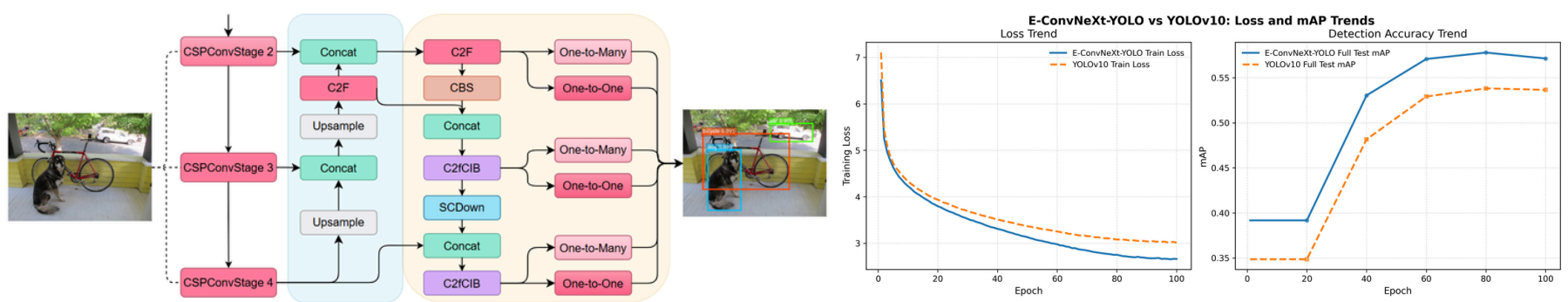


Visual Modules

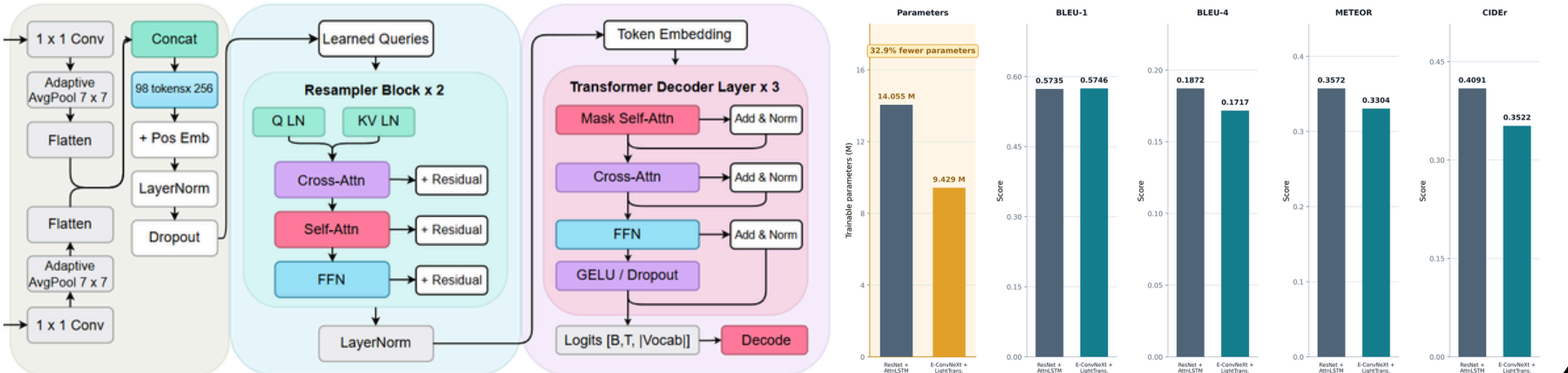
E-ConvNeXt as Lightweight Feature Extractor



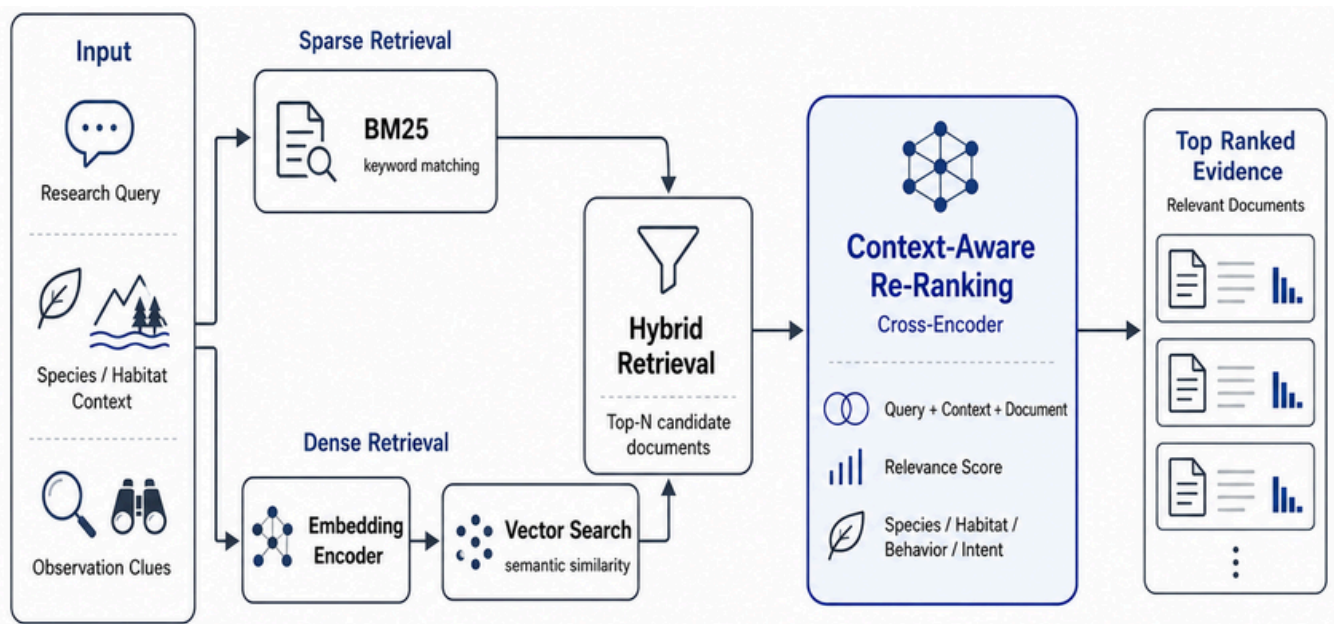
E-ConvNeXt-YOLO for Object Detection



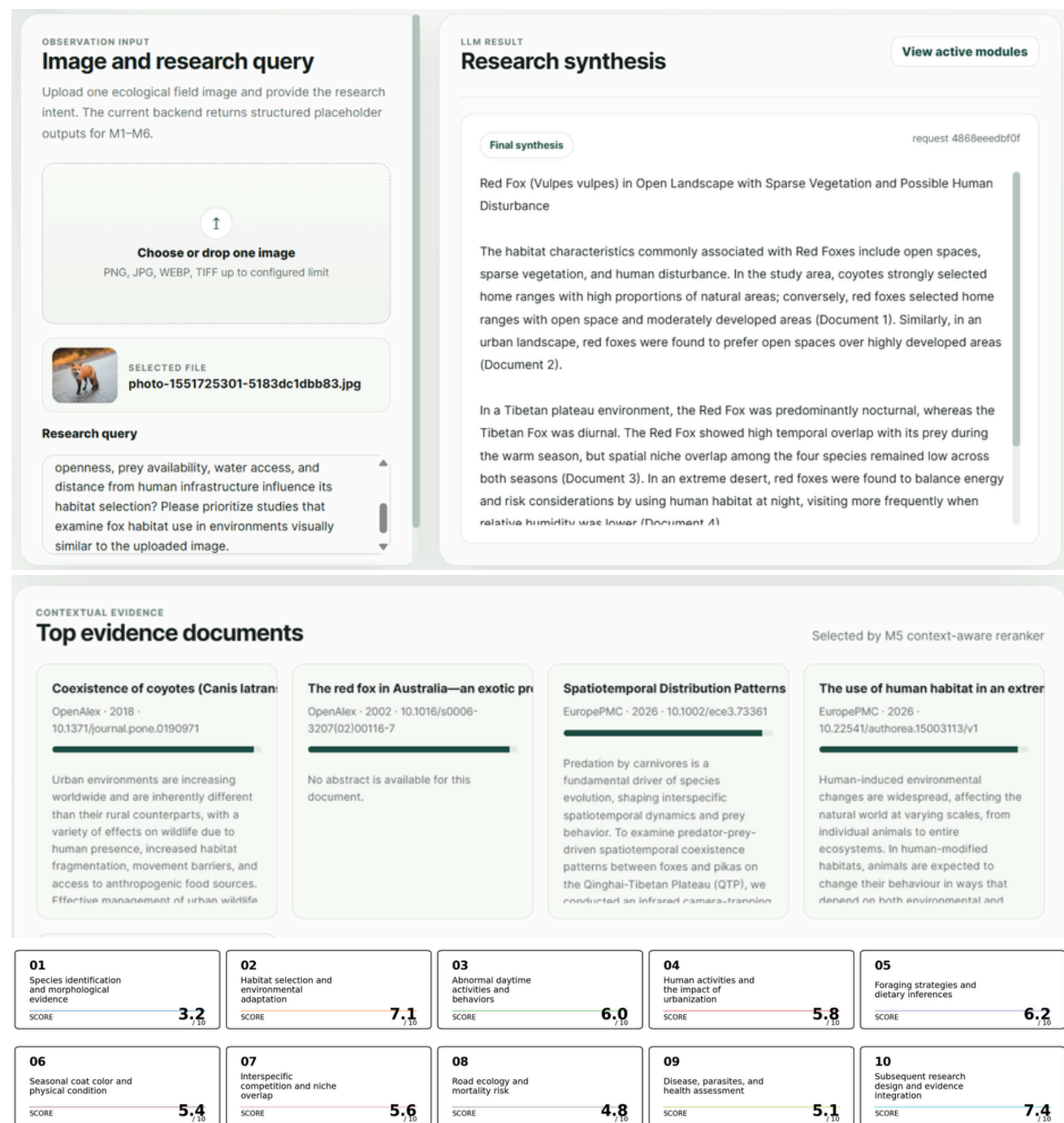
E-ConvNeXt-Transformer for Image Captioning



Literature Retrieval



Demostration



Conclusion

- 1 | E pluribus turns field observations into research-ready evidence by connecting images, literature, and local knowledge.
- 2 | Its key technical value is selecting the right evidence, not more evidence, through context-aware reranking.
- 3 | The project reframes ecological AI from species recognition into a knowledge system for clearer questions and more reliable insight.

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